



Society of Petroleum Engineers

SPE-229966-MS

Successful Implementation of a Microemulsion Surfactant in Solvent Systems Applied During Matrix Stimulation Treatments to Improve Condensate Banking Removal in Gas-Condensate Wells in Mexico

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229966-MS](https://doi.org/10.2118/229966-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

This paper presents the development and field application of an enhanced stimulation methodology for gas and condensate wells in the I Field. The approach was formulated after a detailed evaluation of current reservoir conditions, with the primary objective to address key factors that contribute to formation damage associated with retrograde condensation. The stimulation design, organic in nature, incorporated a next-generation surfactant additive engineered to promote fluid mobility through the formation of a thermodynamically stable microemulsion. Field implementation demonstrated the technique's potential to mobilize condensate banks in the near-wellbore (NWB) region and mitigate their detrimental impact on well deliverability.

Initial matrix stimulation treatments in the field were predominantly acid-based and delivered favorable outcomes during the early stages of field development. These treatments removed NWB damage through acidizing and created new flow pathways, such as wormholes, that helped improve conductivity and reduced the pressure differential between the reservoir and the NWB region. However, progressive reservoir depletion over time led to reduced bottomhole pressure. In several instances, post-treatment pressure approached or fell below the dewpoint of the reservoir fluid, limiting the effectiveness of conventional acid-based techniques.

To overcome this challenge, a customized solvent formulation was developed that lowers surface tension at the gas-condensate interface. The objective was to partially solubilize enriched liquid hydrocarbons around the wellbore and revaporize them into the gas phase. The solvent system, augmented with microemulsified surfactant additives, also reduced interfacial tension (IFT) between water and condensate, which helped improve the mobility of both phases. This contributed to significantly reduced fluid cleanup times and improved post-treatment flowback performance.

The findings underscore the advantages of advanced surfactant additives in solvent- and acid-based stimulation systems. Fundamental benefits observed include:

- Effective mitigation of retrograde condensation-related damage

- Stabilized post-treatment wellhead pressure, which suggests delayed condensate accumulation
- Reduced post-treatment fluid recovery time

Additional stimulation treatment results presented in this publication build upon prior work (de-Antunano-Munoz, Y., et al, 2025), in which stimulation chemistry and laboratory screening methodology for a microemulsion-enabled solvent system are detailed. Production results from these additional wells demonstrate the potential to perform successful stimulation interventions through customized chemistry and pump schedules designed for the specific requirements of the reservoir.

Significance of Gas-Condensate Reservoirs in Mexico

Natural gas is widely recognized as a transitional energy source because of its comparatively lower emissions profile. When combusted for energy, natural gas produces significantly fewer air pollutants, such as carbon dioxide, than coal or petroleum-derived fuels. Per the US Energy Information Administration (EIA 2024) and the Environmental Protection Agency (EPA 2023), natural gas combustion results in 50 to 60% lower CO₂ emissions relative to coal for equivalent energy output.

Because of its cleaner environmental footprint, the development of gas-condensate reservoirs holds greater importance in the global energy landscape. Major fields such as the Arun (Indonesia), Cupiagua (Colombia), Karachaganak (Kazakhstan), North Field (Qatar), and Shtokmanovskoye (Russia) have played pivotal roles in the supply of this energy source (Miller et al. 2010). More recently, a significant gas-condensate reservoir has been identified in Mexico, with substantial production potential (Cruz Luque et al. 2016). Martinez Medrano et al. (2024) documented the successful completion of a gas-condensate well in Mexico in 2017 that targeted a reservoir with a 41° API condensate and gas trapped in carbonate formations composed of fractured limestones and calcite-rich banks with microvugs. In 2018, a reassessment of the reservoir's prospective (P3) reserves estimated 1,314 MMboe, with 8,206 Bcf of original gas in place and 1,002 million barrels of condensate.

After nearly a decade of continuous production, a decline in output was observed in multiple wells in the field, primarily attributed to condensate dropout caused by pressure losses in the NWB region. A laboratory investigation that identified a solvent-based stimulation fluid formulated with a microemulsion surfactant blend, optimized to act as a flowback aid (Khamatnurova et al. 2019), is presented. The optimized system was subsequently applied in six gas-condensate wells in Mexico, with treatment outcomes discussed herein.

Condensate Banking and Remediation Techniques

Condensate banking in wet gas reservoirs form when pressure near the wellbore falls below the dewpoint of the reservoir fluid, which causes liquid hydrocarbons to condense from the gas phase. Fig. 1 illustrates the phase behavior that causes condensate blockage. In the deeper formation, where the average reservoir pressure (P_{ave}) exceeds the bottomhole flowing pressure (P_{wf}), the fluid remains entirely in the gas phase. However, as gas flows toward the wellbore and the pressure approaches the dewpoint, heavier hydrocarbon fractions condense and eventually form a liquid bank once the P_{wf} drops below the dewpoint.

This condensate bank significantly reduces the relative permeability to gas and impairs productivity (Towler 2002). Fig. 1 shows that not all perforations maintain communication with the wellbore, which further increases production losses (Al-Yami et al. 2013). The severity of condensate banking depends on several reservoir and fluid parameters, such as permeability, porosity, hydrocarbon composition (particularly methane content), and reservoir temperature. However, high production rates that create sharp pressure drop in the NWB region often accelerate this effect. Various remediation strategies have been developed to counteract condensate banking, such as pressure maintenance techniques, productivity enhancement treatments, and chemical injection. Notable among these are surfactant-based solvent systems that reduce

IFT and mobilize trapped liquids (Bennion et al. 2001; Kumar et al. 2006; Ahmadi et al. 2011; Sayed and Al-Muntasheri 2014). This study contributes to this body of work through the presentation of the field-scale application of a microemulsion-enhanced solvent system designed to address condensate blockage and help improve flow efficiency in a prominent Mexican gas-condensate reservoir.

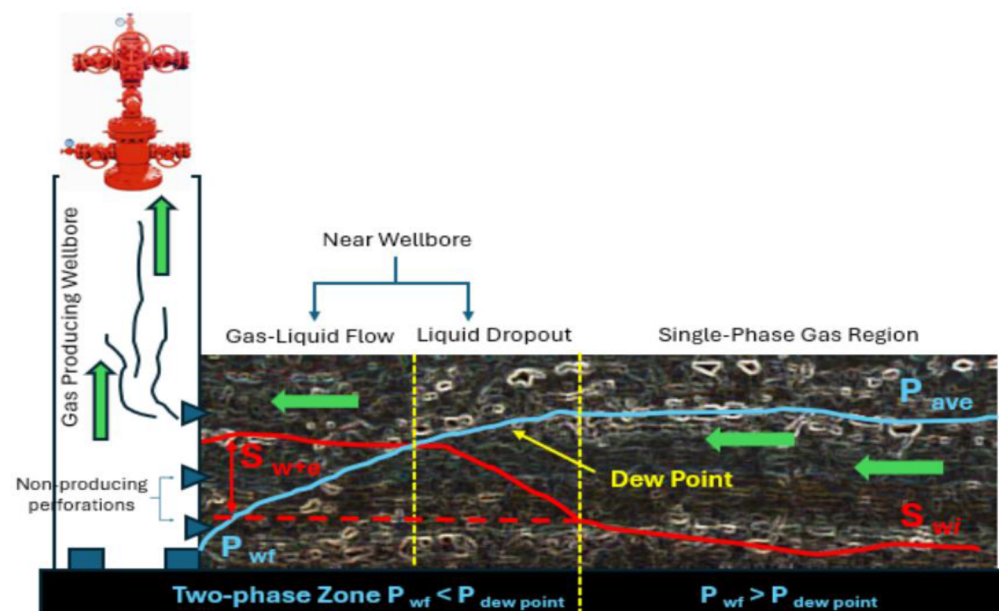


Figure 1—Multiphase flow behavior in a wet gas well. As the bottomhole flow pressure (P_{wf}) falls below the dewpoint, the effective liquid saturation (S_{w+e}) in the NWB zone increases, which reduces gas relative permeability. This pressure drop initiates the condensation of heavier hydrocarbon components from the gas phase to form a liquid-rich region that adversely affects gas production (Al-Yami et al. 2013).

Fig. 2 depicts a summary of field-proven methods for mitigation of NWB condensate and water blockage in gas-condensate reservoirs. A range of remediation techniques has been applied to address hydrocarbon condensate banking and water blockage in the NWB region of the gas-condensate wells (Sayed and Al-Muntasheri 2016).

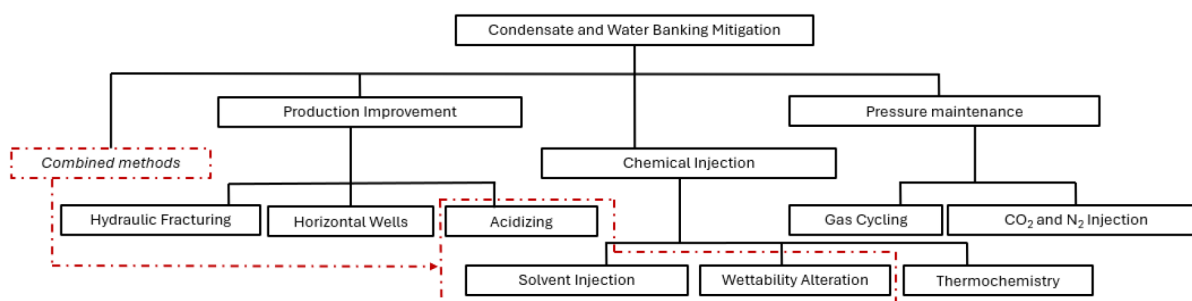


Figure 2—Overview of strategies to help prevent and mitigate condensate banking (Sayed and Al-Muntasheri 2016). The red dashed lines highlight the specific methods applied in this study to help eliminate condensate accumulation and help improve hydrocarbon recovery in a gas-condensate well in Mexico.

To counteract bottomhole flow pressure that falls below the dewpoint, energized fluid (typically CO_2 or N_2) have been injected into the formation to sustain a pressure profile that inhibits liquid dropout (Sharma et al. 2018). Another widely adopted strategy involves drilling horizontal wells in place of vertical ones because the increased reservoir contact area enables a more uniform pressure gradient in the pay zone and reduces the severity of localized pressure drops near the wellbore.

In addition to horizontal well architecture, hydraulic fracturing is frequently employed in gas-condensate wells to help improve formation permeability and conductivity. This approach aids gas mobility under reduced drawdown conditions, which helps maintain the bottomhole pressure above the dewpoint and mitigates condensate dropout.

More recently, thermochemical treatments have regained interest to revaporize liquid hydrocarbons. The treatments induce in-situ chemical reactions that increase temperature and pressure, which help improve hydrocarbon recovery (Hassan et al. 2019). As indicated by the red dashed lines in Fig. 2, the present study focuses on a field application in Mexico that used a two-stage intervention. The first stage applied solvent injection with a surfactant-based flowback aid to dissolve and mobilize NWB condensate accumulation (Du et al. 2000). The second stage applied a targeted acid stimulation treatment to restore pressure communication and boost drawdown efficiency in gas-condensate wells.

Flowback Aids for Gas-Condensate Remediation

The laboratory development and field application of a solvent-based preflush with a surfactant-based flowback aid that is designed to enhance the removal of liquid hydrocarbon dropout in the NWB region of gas-condensate reservoirs are discussed. Flowback aids, typically surfactant blends, are conventionally applied during hydraulic fractures to facilitate the recovery of fracture fluid from stimulated formations (Leal et al. 2013; Xu 2013; Kim et al. 2016; Yue et al. 2016; Khamaturova et al. 2019; Recio et al. 2019).

After the solvent injection, an acidizing treatment is applied to help improve formation permeability and reduce pressure differential between the wellbore and reservoir (Al-Sabea et al. 2022; Lugo et al. 2023). These nonreactive, non-energized chemical injection systems operate through several mechanisms: reduction of air/water and oil/water IFT (Grenoble and Trabelsi 2021), wettability alteration at the fluid-rock interface (Gilani et al. 2011; Wang et al. 2015; Kazemi et al. 2024), increased detergency, improved dispersion, demulsification capability, and thermal stability.

Kazemi et al. (2024) conducted a comprehensive review that emphasized the critical role of wettability modification in both prevention and remediation of banking. Their results indicate a predominant research focus on fluorinated polymers, surfactants (Fahimpour et al. 2012; Karandish et al. 2015), and alkyl fluoride-modified nanoparticles. However, relatively few studies have examined the synergistic use of nonionic surfactants with nanoparticles. Franco-Aguirre et al. (2018) explored this combination, while Garzon et al. (2006) reported that nonionic surfactants negatively impacted hydrocarbon recovery in core flow studies that simulated gas-condensate conditions.

In the context of microemulsion-based flowback aids for wettability modification, Rostami et al. (2016) reported that microemulsions exhibited low IFT values and intermediate contact angles in laboratory tests, which translated into reduced capillary pressures. Core flood experiments using Berea sandstone cores demonstrated improved effective gas permeability and longer-lasting treatment efficacy, results comparable to those obtained using a fluoropolymer-mutual solvent formulation.

More recently, Saini et al. (2024) evaluated surfactant blends formulated as flowback aids to help improve acid stimulation in calcite core systems under simulated gas-condensate conditions. Their results highlighted the superior performance of microemulsion-based additives that remained stable in spent acid, formed no emulsions with condensate hydrocarbons, improved fluid displacement in column tests, and improved regained permeability to nitrogen gas in core flow experiments.

Because of the limited published data on the use of flowback aids specifically designed for solvent-based condensate remediation, current work contributes a novel laboratory-to-field case study. The approach resulted in improved production outcomes from wet gas wells in Mexico. Table 1 shows several surfactant-based formulations that were identified as high-performance flowback aids suitable for gas-condensate remediation. These candidates were evaluated for compatibility with produced water at reservoir temperature, their effectiveness in the prevention of emulsion formation, and their ability to facilitate rapid

displacement of stimulation fluid with hydrocarbons in column sandpack tests (Himes et al. 2017; Nelson et al. 2018; Recio et al. 2018, 2019).

Table 1—Flowback aid information.

Flowback Aids	Description	Relative Solubility Number
FA-A	Blend of surfactants formulated as a microemulsion	13
FA-B	Conventional anionic surfactant	
FA-C	Mixed anionic and nonionic surfactant blends with a demulsification capability	16
FA-D	Nonionic APE-based blend of surfactants with a demulsification capability	12
FA-E	Mixed cationic surfactant with a demulsification capability	15

Experimental Materials

The experimental program used a series of solvent-surfactant formulations and crude oil samples to evaluate dissolution behavior and flowback aid performance. Tables 2 and 3 summarize system components and crude oil characterization.

Table 2—System component descriptions, KB (Kauri-Butanol value)

System Components	Solvent A	Solvent B
Solvent	Solvent with a KB value of approximately 98	Solvent with a KB value of approximately 90
Cosolvent	Solvent with a KB value greater than 200	
Crude oil stability enhancer	Solvent with a KB value of approximately 100	
Dispersant	Salt, anionic surfactant	
Surfactant	Surfactant	
Alcohol	Alcohol	

Table 3—Crude oil characterization.

Crude Oil	API – Gravity	Density	Asphaltene %	Paraffin %	Water %	Solids %
Well B	38.8	0.83	0.63	13.24	0.25	0.01
Well E	42	0.812	0.28	2.99	9	0.01
Well F	37.6	0.836	0.34	5.97	0.2	0.01

Laboratory Results and Discussion

Evaluation of Flowback Aids

Surfactant blends formulated as flowback aids can serve as performance-enhancement additives in stimulation fluid, particularly when applied to hydrocarbon deposition in the NWB region. Solvent-based pre-flush fluid, which are often highly volatile, are typically injected into the formation before the primary stimulation treatment to dissolve or displace accumulated organic deposits such as asphaltenes or paraffins.

In gas-condensate reservoirs, condensate banks form when the bottomhole flow pressure falls below the dewpoint, which causes heavier hydrocarbons to condense from the gas phase. These condensed liquids significantly impair relative permeability to gas and reduce well productivity. A laboratory program

addressed this issue; it identified optimal flowback aid and solvent-based fluid systems, which remediate condensate banking and render the formation surfaces water-wet post-treatment.

The laboratory study used formation-representative materials such as quartz- and feldspar-dominated minerals, produced water with total dissolved solids (TDS) greater than 9,000 ppm, and a representative hydrocarbon with an API gravity greater than 60°. Table 1 summarizes the flowback aid candidates evaluated for compatibility with condensate and effectiveness in the improvement of mobility.

Fluid compatibility results (Fig. 3) show that flowback aids with anionic character (FA-B and FA-C) exhibited poor compatibility with the high-salinity formation brine. Test conditions evidenced this through the appearance of precipitates and emulsions. In contrast, the nonionic alkylphenol ethoxylate (APE)-based flowback aid (FA-D) reached its cloud point at reservoir temperature, as indicated by the transition to an opaque appearance.

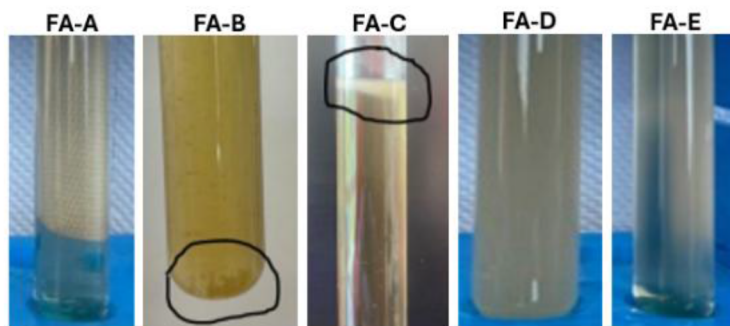


Figure 3—Compatibility tests of formation brine with various flowback-aid formulations at reservoir temperature (200°F). Flowback aids FA-A (microemulsion based) and FA-E (with mixed cationic surfactant components) exhibited good compatibility, as indicated by low turbidity and the absence of precipitates. In contrast, FA-B and FA-C, both anionic in nature, exhibited poor compatibility, characterized by visible precipitation.

Of the systems tested, FA-A and FA-E demonstrated superior thermal stability and compatibility, with a translucent phase maintained throughout the heat-up process. These observations suggest that FA-A and FA-E are better suited for deployment in high-salinity, high-temperature gas-condensate environments.

In addition to an evaluation of the impact of different flowback aids on fluid stability (Fig. 3), it is critical to ensure that their introduction does not promote formation or persistence of emulsions. Fig. 4 presents the emulsion break-time test results for various flowback aid formulations. Based on the formulation properties summarized in Table 1, flowback aid FA-E, characterized by high detergency and effective demulsification capabilities, exhibited the fastest emulsion resolution, with the ability to achieve 100% phase separation within five minutes. FA-A also demonstrated a rapid break time of five minutes, though it did not achieve complete separation within 10 minutes at ambient conditions. While FA-D reached full separation after 10 minutes at ambient temperature, it exhibited premature clouding during compatibility tests, which indicated potential instability at reservoir conditions. Formulations FA-B and FA-C failed to achieve complete separation under the test conditions, which may be attributed to their anionic surfactant composition and corresponding interfacial behavior.

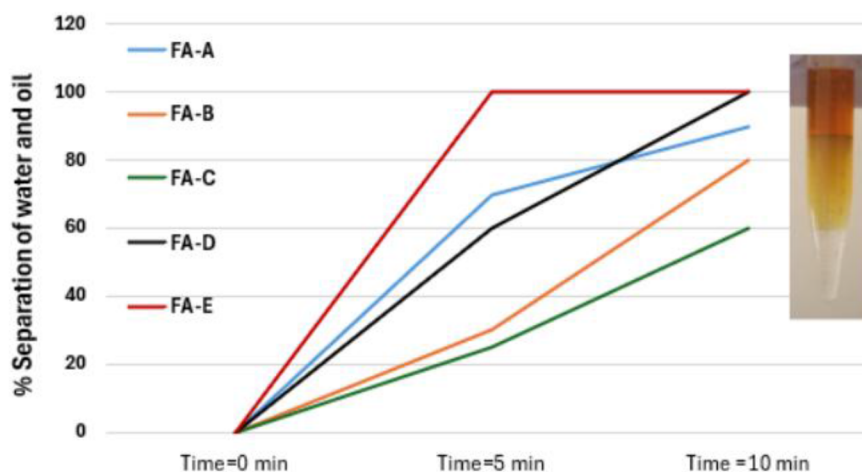


Figure 4—Evaluation of fluid compatibility based on the time it takes for emulsions to separate. The extent of separation between hydrocarbon and water-based fluid was monitored over time. Flowback acid FA-E showed the most effective performance, while FA-C was less effective, potentially because of its anionic properties.

Column flow tests provide an integrated platform to evaluate the effectiveness of stimulation-fluid formulations in conjunction with formation materials. This method enables a comprehensive assessment of new chemical additives that help improve hydrocarbon mobility through sandpacks with formation cuttings. As shown in Fig. 5, the control scenario demonstrates that, in the absence of surface-active agents, hydrocarbon cannot penetrate a brine-saturated, cuttings-laden sandpack. This outcome results from the high oil/water IFT, which inhibits the displacement of brine with the hydrocarbon phase. Introduction of a flowback aid or stimulation-fluid system that reduces IFT is required for the hydrocarbon to overcome capillary barriers and displace the brine.

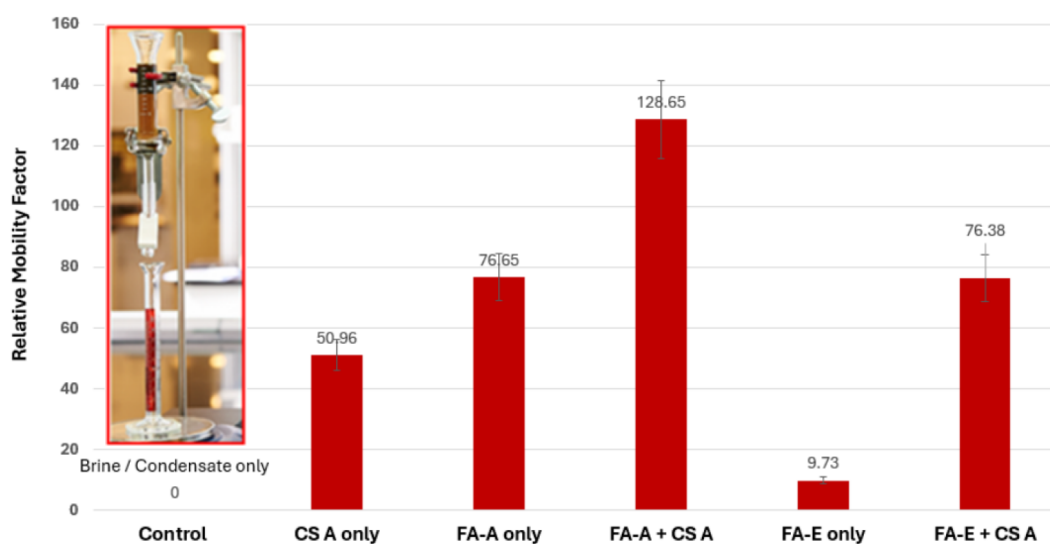


Figure 5—Comparative column flow test results expressed as Relative Mobility Factor, calculated with Eq. 1. Control trials confirm that the placement of hydrocarbons above a sandpack fully saturated with formation brine does not result in condensate infiltration or fluid displacement for at least two hours. The crude oil remains immobile within the column until a surface-active agent, such as a cosolvent or surfactant, alters interfacial properties.

To better represent reservoir conditions and the role of wettability in capillary-pressure reduction, the test matrix included 30% formation cuttings into the sandpack. Although wettability contributes to fluid imbibition, extensive column flow tests consistently showed that oil/water IFT remains the dominant factor that influences the hydrocarbon's relative mobility through the porous media. Fig. 5 further confirms that no hydrocarbon breakthrough occurs in the absence of a cosolvent (CS) or flowback aid (e.g., FA-A).

Based on prior fluid compatibility evaluations at reservoir temperature and emulsion break-time analyses, flowback aids FA-A and FA-E were selected for combination with a cosolvent. This formulation was tested to determine its effectiveness to help improve the mobility of a light, condensate-prone crude oil (API gravity greater than 60) through a sandpack saturated with stimulation fluid.

Fig. 5 shows a summary of the experimental results that assess hydrocarbon displacement through a sandpack column using combined flowback aid and cosolvent systems. Column flow tests followed Procedure A, in which the mass of displaced fluid over time was recorded. This value, multiplied by the fraction of pore volume displaced (2.5 mL), was then applied to Eq. 1 to calculate the Relative Mobility Factor (RMF).

$$\text{Relative Mobility Factor (RMF)} = \frac{\text{Final mass of fluid displaced from column}}{\text{Breakthrough time (BTT)}} \times \% \text{ of Sandpack Pore Volume Displaced} \quad \text{Eq. 1}$$

Flowback aid FA-A is a microemulsion-based surfactant blend. Historical interfacial tension measurements using a broad range of crude oils indicate that this formulation effectively reduces oil/water IFT to less than 0.1 mN/m. CS-A is a high-polarity cosolvent widely adopted in the industry for its dual ability to solubilize both polar and nonpolar hydrocarbon constituents.

Flowback aid FA-E is a conventional surfactant system that incorporates a high-detergency, cationic surfactant component along with chemical intermediates that function as demulsifiers. Although FA-E includes additional formulation complexity, experimental results demonstrate that the combination of FA-A with CS-A yields a superior synergistic effect that greatly improves hydrocarbon mobility and displacement through the sandpack.

Impact of Flowback Aid and Solvent Selection on Stimulation Fluid for Condensate Remediation

The gas-condensate field in Mexico, discovered in 2017, is estimated to produce as much as 80,000 and 780 Mcf/D of gas. Situated in a basin composed of shaly limestones and Upper Miocene shales, this field represents one of the most significant onshore hydrocarbon discoveries in the country. Because of its strategic importance to Mexico's energy development, optimization of production performance in the asset is critical.

As the field matures, reduced reservoir pressure creates operational challenges typical of wet gas systems. When bottomhole pressure falls below the dewpoint, heavier hydrocarbon fractions condense and accumulate in the NWB region, which impairs permeability and reduces gas flow, a phenomenon known as "condensate banking." To mitigate this damage and restore production, stimulation fluid packages are formulated to mobilize trapped condensate and render the formation surfaces water-wet. A laboratory-based evaluation that identified the optimal stimulation fluid formulation for condensate remediation in this field is presented.

Stimulation Fluid Formulation Strategy

Initial efforts to design an effective remediation fluid focused on the evaluation of various stimulation formulations that incorporated various flowback aids and solvents to help improve fluid compatibility and operational resilience (Table 2). These formulations typically included a combination of:

- High- or low-volatility solvent
- High-polarity cosolvent
- Crude oil stability enhancer

- Linear alkyl ammonium salt dispersant
- Oil-soluble surfactant
- Linear alcohol
- Flowback aid

The component selection followed two primary criteria: (1) stability in the presence of high salinity produced brines and (2) solubility in a broad polarity range of condensate constituents.

Crude oil samples from multiple wells in the gas-condensate field were characterized (Table 7), and the results indicate that condensate dropout damage is primarily paraffinic in nature. Because paraffins and asphaltenes differ markedly in polarity, stimulation fluid solvents had to demonstrate compatibility with both high- and low-polarity hydrocarbon components.

To address these polarity-related challenges, the formulation included a crude oil stability enhancer to minimize potential component separation downhole. A linear alkyl ammonium salt dispersant was selected for its dual role: stabilization of the formulation and dispersion of polar hydrocarbon fractions (e.g., asphaltenes). In addition, a hydrocarbon-soluble, nonionic surfactant was used to reduce emulsion formation through competitive adsorption at the oil/water interface and displace native surface-active crude oil components.

A key component of the system is the flowback aid, which plays a critical role in the reduction of oil/water IFT and the promotion of water-wet conditions in the formation, both essential for efficient hydrocarbon displacement and improved injectivity.

Laboratory Evaluation and Results

The experimental campaign to identify the most effective stimulation fluid began with an assessment of surface-tension reduction in various formulations. Surface tension was measured using the Du Noüy ring method, with results summarized in Fig. 6.

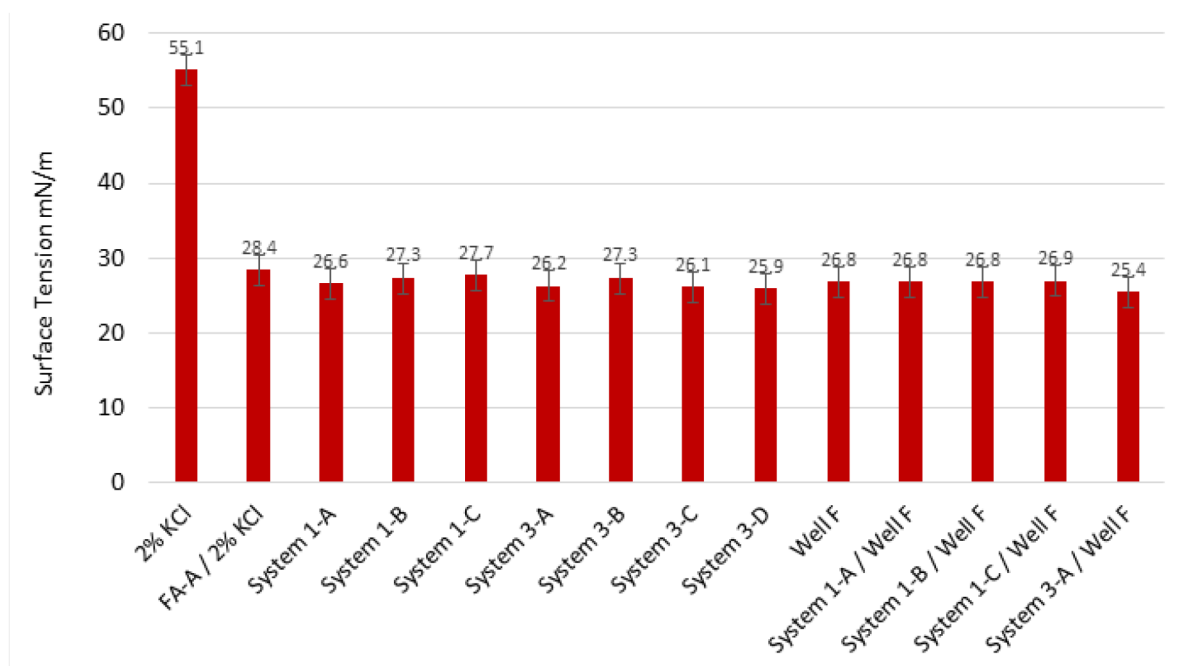


Figure 6—Surface-tension analysis at the air/fluid interface: compatibility assessment of hydrocarbon solubilization systems with formation brine through the Du Noüy ring technique.

The introduction of 0.4 wt% flowback aid into brine resulted in a marked reduction in surface tension, which demonstrated the additive's effectiveness. However, comparative analysis of the full fluid systems indicated that inclusion of the flowback aid did not significantly destabilize or improve surface-tension performance across the board. Specifically:

- Addition of flowback aid to base fluid (e.g., 2% KCl) produced less than 5% variation in surface tension compared with the full stimulation system (e.g., System 1-A, [Table 3](#)), which suggests negligible formulation impact at the laboratory scale and reliable expectations for field performance.
- Incorporation of linear alcohol (as observed in the progression from System 1-C to System 3-D, [Table 4](#)) produced the most pronounced improvement in surface-tension reduction, a parameter likely to correlate with stronger field performance.

Table 4—Flash points of various stimulation systems to highlight field deployment safety.

System	Flash Point °C
System 1-A	47.4
System 1-B	57.7
System 6	50.5
System 2-A	52
System 2-B	52
System 7	> 200

However, when IFT was measured, a more critical indicator of hydrocarbon mobility, the differences between formulations became more pronounced. These results underscore the importance of IFT reduction, in addition to surface tension, in the evaluation of stimulation fluid for condensate-remediation applications.

In addition to surface-tension analysis, oil/water IFT measurements were conducted to evaluate the compatibility and effectiveness of stimulation fluid systems in the remediation of formation damage caused by condensate dropout in wet-gas wells. These measurements provide insight into the ability of a formulation to alter wettability and help improve fluid mobility in the reservoir. Because of the critical role of capillary pressure in film formation and detergency (both functions of surface wettability), the target performance threshold for stimulation fluid is an oil/water IFT lower than 1 mN/m. This benchmark reflects values relevant to both the Frumkin-Derjaguin equation, which describes film-tension gradients, and the Young-Laplace equation, which estimates capillary pressure. Together, these equations help predict a formulation's impact on the dynamics of spread and wettability alteration ([Kondiparty et al. 2011](#)).

IFT measurements used the Du Noüy ring method, and [Fig. 7](#) shows the results. For dual-component systems (e.g., System 1-A with brine), fluids were mixed at a 50:50 volumetric ratio. In tri-component systems, where hydrocarbon, stimulation fluid, and brine were present, measurements used a 25:25:50 ratio, respectively. Control tests also measured Well F and Well E condensates in both brine and fresh water. Data in [Fig. 7](#) show that all stimulation fluid tested exhibited significant reductions in IFT when compared to the control scenarios. Notably, the Well E condensate sample demonstrated a higher sensitivity to brine than Well F, as evidenced by more pronounced IFT changes in the presence of salinity.

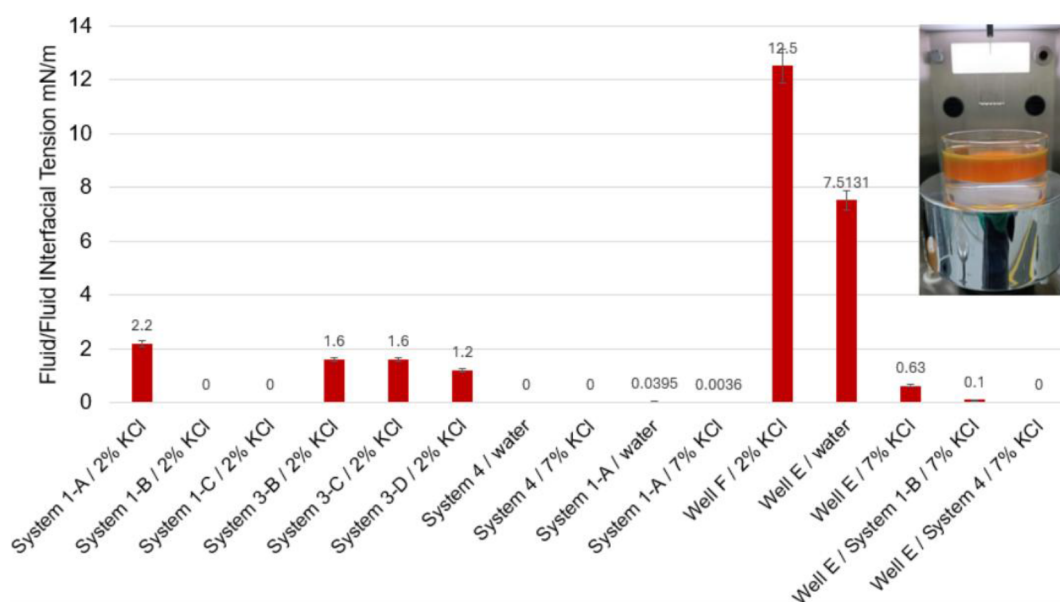


Figure 7—Oil/Water IFT analysis: evaluation of interactions between hydrocarbon solubilization systems and formation fluid. Test mixtures used system-to-brine ratios of 50:50 and system-to-crude-to-brine ratios of 25:25:50, respectively.

The column-flow test methodology integrates all relevant formation fluid and solids that a stimulation fluid system could encounter under downhole conditions (refer to Procedure B). The sandpack used in the column is composed of a blend of proppant and formation cuttings, milled to a uniform mesh size to help ensure consistency and repeatability of results. Before the test, this mixture was conditioned with produced water specific to the formation and then saturated with the selected stimulation fluid formulation. As shown in Fig. 8, Well E crude oil was introduced above the stimulation-fluid-saturated sandpack to initiate hydrocarbon flow through the system.

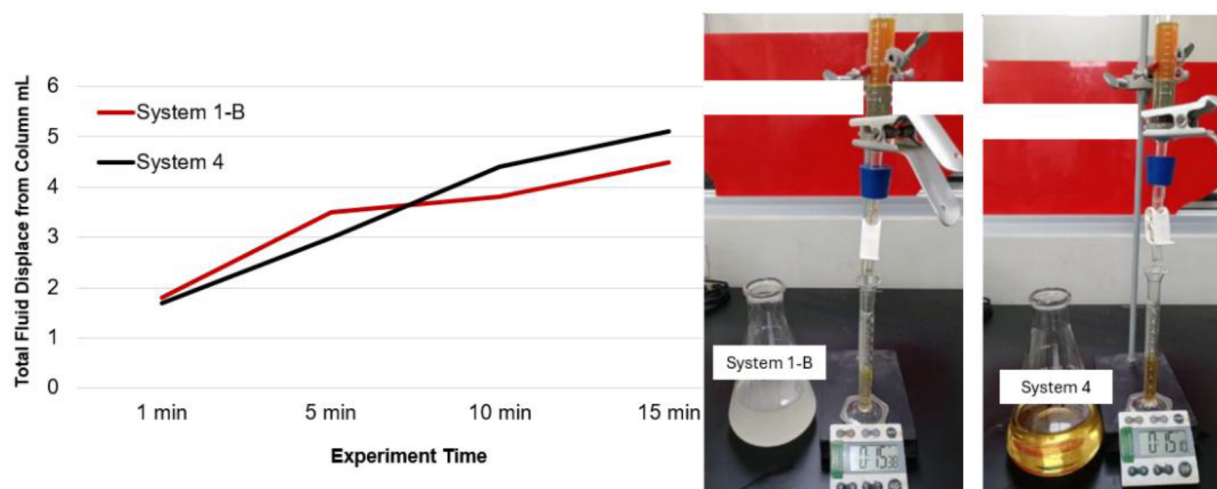


Figure 8—Column flow evaluation: assessment of brine displacement with hydrocarbons and the resultant hydrocarbon flow rate through the sandpack. The pore volumes for sandpacks used in System 1-B and System 4 tests were 4.0 and 3.8 mL, respectively.

Comparative results between System 1-B and System 4 indicate that both systems achieved complete displacement of the sandpack's pore volume within the first 10 minutes of the test. The oil level was maintained at the 15-mL mark, and the effluent volume was measured continuously to assess displacement rate and flow behavior. Performance differences emerged during the test. While both systems initially performed comparably, System 1-B demonstrated superior displacement efficiency during the first eight

minutes. This initial advantage likely resulted from the flowback aid in System 1-B being present at concentrations greater than its critical micelle concentration (CMC), which effectively stabilized the oil/water interface and promoted efficient displacement. However, after the eight-minute mark, performance declined, which suggests that the concentration of flowback aid in System 1-B dropped below the CMC because of adsorption onto formation materials, a behavior previously reported in the literature (Zheng and Rao 2011). As a result, the stabilization of the oil/water interface diminished, and displacement efficiency decreased thereafter.

In contrast, System 4, formulated without a flowback aid, maintained superior displacement performance throughout the test. This sustained performance can be attributed to the higher overall concentrations of crude oil solubility enhancer and surfactant in the formulation (Table 5), which collectively supported continuous crude oil mobilization and displacement through the sandpack (Frost et al. 2008). These observations align with the IFT measurements with Well E crude oil, where System 4 achieved a near-zero IFT, compared to approximately 0.1 mN/m for System 1-B. The enhanced IFT reduction capability of System 4 further validates its superior performance in the column flow test and its potential field applicability for remediation of condensate banking damage.

Table 5—Actual pump schedule implemented for the execution of the primary solvent treatment.

Stage	Fluid Name	Stage Volume	Pumping Rate	Stage Time (min)
Main solvent stage	System 1-B	63 bbl	4 bbl/min	15.7
Main solvent stage	System 1-B	126 bbl	5 bbl/min / 105 scm/min	25.2
Main solvent stage	System 1-B	126 bbl	6 bbl/min / 140 scm/min	21.0
Displacement	Nitrogen	9,100 scm	220 scm/min	41.3
Shut in (soaking time)	NA	NA	NA	120.0

In addition to laboratory performance, the flash point of stimulation fluid formulation is a critical parameter for safe management and field deployment. As summarized in 4, flash-point measurements serve as key indicators of the volatility and flammability risks associated with solvent-based systems. Among the fully solvent-based formulations tested, System 1-B exhibited the highest flash point, while System 7 showed a lower flash point because of its substantial water content. The relatively high flash point of System 1-B, combined with its demonstrated efficacy, positions this solvent-based stimulation fluid as a suitable candidate for both offshore and onshore well remediation to help provide improved safety without loss of performance.

Based on the results and analysis discussed, as well as the comparable performance of Systems 1-B and 4, System 1-B was selected for a chemical stimulation campaign designed to mitigate condensate banking damage and accelerate production recovery in multiple wells within the Mexican gas-condensate field. The subsequent sections provide a detailed overview of the well selection criteria, deployment methodology, and the production outcomes observed after the implementation of Stimulation Fluid System 1-B in the field.

Field Implementation in Matrix Stimulation

Field Overview

The selected field for application of this stimulation fluid technology is a prolific onshore high-pressure/high-temperature (HP/HT) gas-condensate reservoir recently discovered in Mexico. Initial exploration and development of the Middle Cretaceous (MC) reservoir began in 2017. The reservoir consists predominantly of limestone with variable textures that reflect deposition within a complex reefal environment. Wells

in this field typically reach total measured depths of approximately 6900 m. Initial reservoir pressure exceeded 1200 kg/cm², and bottomhole temperature ranged between 150° and 160°C. After nearly eight years of production, reservoir pressure declined significantly. The MC formation exhibits an average porosity of 5% and extremely low matrix permeability, between 0.01 and 0.05 millidarcies (mD). These characteristics, combined with heterogeneous depositional features, result in significant variability in reservoir performance.

Operational Challenges

Most of the candidate wells for this application are slightly deviated and produce hydrocarbons commercially from a single perforated interval within the MC reservoir. Gas-condensate dropout was observed early in the production life of these wells, even before significant pressure drawdown occurred, an indication of early condensate banking in the NWB region.

Initial matrix stimulation treatments in the field primarily used hydrochloric acid-based systems and yielded favorable results in the early years of field development. These acid treatments effectively removed damage through wormhole creation and restoration of flow pathways, which lowered the pressure differential between the reservoir and wellbore (Lugo et al. 2023). However, as reservoir pressure declined with time, post-treatment bottomhole pressure frequently approached (or fell below) the dewpoint of the reservoir fluid, which significantly reduced treatment effectiveness (Sayed and Al-Muntasheri 2016).

The primary technical challenge was to formulate a solvent-based system capable of surface-tension reduction at the gas-condensate interface to partially solubilize liquid hydrocarbons in the NWB region and promote revaporization into the gas phase (Du et al. 2020). The selected formulation also incorporated microemulsion-based surfactants designed to reduce water condensate IFT and mitigate the capillary forces that hinder hydrocarbon mobility (Rostami et al. 2016).

Well A: Diagnostic Analysis and Treatment Design

A pilot matrix stimulation treatment was executed on Well A in December 2023. The treatment objective was to help improve hydrocarbon production and reduce post-stimulation fluid recovery time. Pretreatment diagnostics included wellhead pressure and flow rate measurements in a range of choke sizes. These data were used to construct a productivity index model that incorporated current reservoir, wellbore, and surface conditions. A sensitivity analysis evaluated the impact of various skin values on production potential.

As part of the treatment design, a pressure buildup (PBU) test was performed to determine bottomhole flow pressure (BHFP). The PBU results confirmed significant NWB formation damage consistent with condensate banking. Fig. 9 presents the interpretation of the PBU analysis and shows the impaired permeability zone around the wellbore. Continuous surface measurements before the treatment revealed unstable wellhead pressure and temperature, which suggested fluctuation in flow conditions likely influenced by two-phase flow dynamics. In addition, pretreatment condensate samples exhibited variation in API gravity, which further confirmed the presence of phase separation and retrograde condensation below the dewpoint.



Solvent Stage – Primary Matrix Treatment. The stimulation treatment for Well A was executed in two stages. The first stage involved the injection of 315 bbl of the solvent-based preflush formulation, designated as System 1-B, under matrix injection conditions at rates between 4 and 6 bbl/min. After the preflush, the treatment was displaced using 9100 standard cubic meters (scm) of nitrogen gas, injected at a rate of 220 scm/min. The final phase of the treatment consisted of the injection of an additional 252 bbl of the solvent formulation, assisted with nitrogen at a reduced rate of 105 scm/min, to help improve fluid placement and penetration into the formation.

Fig. 10 and 5 illustrate the wellhead pressure response and injection rates recorded during the primary solvent injection stage.



Acid Stage – Secondary Matrix Treatment. The second stage of the matrix stimulation treatment involved the injection of 126 bbl of a retarded acid formulation, optimized with flowback aid FA-A to help improve recovery of water-based fluid after the two-stage treatment sequence. The acid volume was delivered in two equal 63-bbl substages, separated with an intermediate solvent spacer to help improve acid diversion and zonal contact.

Fluid were pumped at rates between 5 to 8 bbl/min. The displacement phase used a total of 8100 scm of nitrogen gas, injected at a rate of 240 scm/min, to assist in fluid placement and minimize liquid retention within the formation (Fig. 11 and Table 6).

Table 6—Actual pump schedule implemented for the execution of the acid stage.

Stage	Fluid Name	Stage Volume	Pumping Rate	Stage Time (min)
Acid	Retarded Acid	63 bbl	5 bpm / 125 scm/min	12.6
Solvent	System 1-B	63 bbl	6 bpm / 165 scm/min	10.5
Acid	Retarded Acid	63 bbl	7 bpm / 210 scm/min	9.0
Solvent	System 1-B	31.5 bb	8 bpm / 240 scm/min	3.9
Displacement	Nitrogen	8,100 scm	240 scm/min	33.7

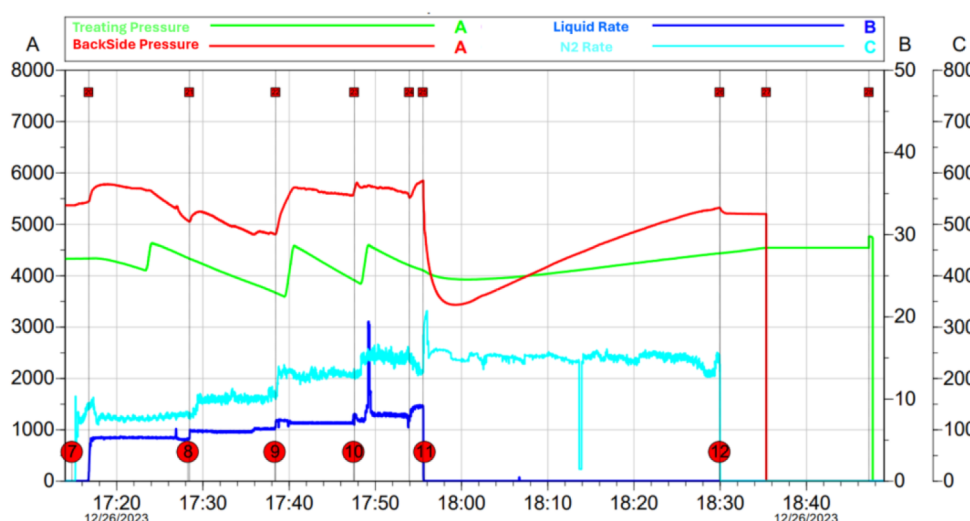


Figure 11—Pump graph: acid stage and nitrogen displacement.

Treatment Results. A total of three matrix stimulation treatments were performed on Well A. The first two, conducted in August 2021 and January 2023, focused on conventional acid stimulation. These treatments used retarded acid formulations designed to generate wormholes, help improve NWB conductivity, and advance reservoir-wellbore communication. Both acid stimulations included a short organic preflush before acid injection.

One year later, after the detailed laboratory and diagnostic work described in the previous sections, a third matrix stimulation was executed; however, this stimulation was focused on condensate blockage remediation. The treatment incorporated an organic solvent system formulated to mobilize and dissolve liquid hydrocarbons condensed in the NWB region. A two-hour soak period followed injection to allow chemical interaction with the formation and maximize the contacted rock surface area.

Comparative production data indicate a significant improvement after the third stimulation. Table 7 shows that oil production measured at a fixed 32/64-in. choke size increased by 350% relative to previous stimulation treatments, which evidently demonstrates the effectiveness of the solvent-based approach for remediation of condensate banking in this reservoir.

Table 7—Well comparison between treatment simulations in Well A.

Well	Operation	Date	Choke Size	Before Intervention		After Intervention		Incremental Qo (B/D)
				Qo (B/D)	Ptp (kg/cm ²)	Qo (B/D)	Ptp (kg/cm ²)	
Well A	Stim	August 2021	0.5	1,666	454.6	1,870	503.9	204
Well A	Stim	January 2023	0.5	2,099	235.7	2,224	247.5	125
Well A	Stim	December 2023	0.5	1,583	260.6	2,021	287.8	438

As illustrated in Fig. 12, the initial interventions were centered on **acid-based stimulation**, which produced a notable short-term increase in oil production. However, production rates gradually declined and eventually returned to pretreatment levels within approximately three months. In contrast, when the matrix stimulation strategy shifted to **organic solvent-based treatments** designed for **condensate blockage removal**, a more substantial and sustained increase in oil production was observed. The production enhancement achieved with the organic treatment persisted for up to **one year post-intervention**, which demonstrates a significantly improved and longer-term impact compared with the earlier acid-only treatments.

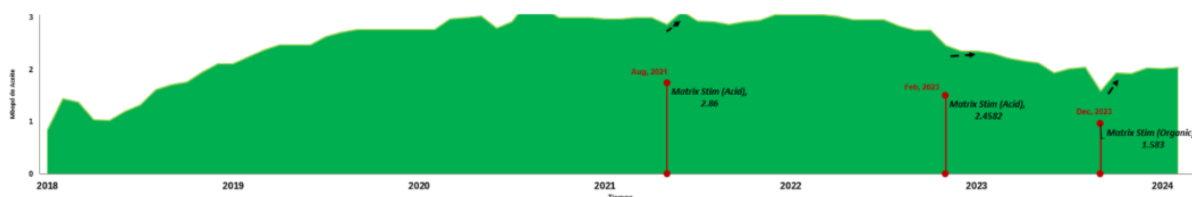


Figure 12—Oil production history for Well A.

Based on the field's production performance, the majority of wells appear affected by condensate blockage, as indicated by accelerated wellhead pressure decline and lower production rates. To validate this hypothesis, each candidate well underwent comprehensive evaluation that included PBU tests, nodal analysis calibration, and produced fluid characterization. Wells that exhibited formation damage consistent with condensate accumulation, evidenced by positive skin values and impaired flow near the wellbore, were selected for treatment using an organic solvent system as the primary remediation approach. Table 8 and 9 summarize the outcomes of these interventions and highlight significant improvements in both oil and gas production in all treated wells.

Table 8—Summary results after well interventions.

Well	Operation	Date	Choke Size	Before Intervention			After Intervention			Incremental Qo (B/D)
				Qo (B/D)	Qg (MMscfd)	Ptp (kg/cm ²)	Qo (B/D)	Qg (MMscfd)	Ptp (kg/cm ²)	
Well A	Stim	12/26/2023	0.5	1,583	13.33	260.6	2,021	14.01	287.8	438
Well B	Stim	3/12/2024	0.75	2,278	19.32	164.6	2,930.0	23.3	214.5	652
Well C	Stim	3/22/2024	0.5625	1,376	12.7	182.8	1,508.0	12.8	186.9	132
Well D	Stim	5/13/2024	1	4,037	32.33	179.2	4,744.0	37.7	208.3	707
Well E	Stim	6/18/2024	0.75	2,383	17.5	168.6	2,915.0	23.4	223.0	532
Well F	Stim	2/24/2025	1	1,055	7.69	94	3,750	28.07	147	2,650
Well G	Stim	3/12/2025	0.5	820	7.81	147	1,730	14.4	283	910
Well H	Stim	4/23/2025	1	3,690	35.36	175	4,306	37.1	187	616
Well I	Stim	5/06/2025	1	1,468	12.95	97	3,129	27.0	147	1,661
Well J	Stim	5/17/2025	0.75	2,010	18.2	142	2,236	20.0	162	226
Well K	Stim	6/13/2025	0.75	1,748	18.8	148.6	2,178	21.6	174.5	430
Well L	Stim	6/21/2025	1.4	4,090	39.77	139	5,099	48.8	139	1,009

Table 9—Summary of values for condensate-to-gas ratios (CGR) observed in wells before and after interventions.

Well	Before Intervention CGR (bbl/MMscf)	After Intervention CGR (bbl/MMscf)	Difference CGR (bbl/MMscf)
Well A	119	144	25
Well B	118	126	8
Well C	108	118	10
Well D	125	126	1
Well E	136	125	-11
Well F	137	132	-5
Well G	105	120	15
Well H	104	116	12
Well I	113	116	3
Well J	110	112	2
Well K	93	101	8
Well L	103	105	2

The incorporation of specialized additives into the primary treatment system demonstrated a significant reduction in post-treatment cleanup times compared with conventional interventions. This development helped improve operational efficiency, which reduced intervention duration and overall treatment cost.

Conclusions

The combination of a flowback aid with a solvent-based stimulation fluid system helped improve the surface-tension reduction potential of the treatment fluid, a critical parameter for removal of phase blockages in gas-producing wells.

Laboratory investigations identified a microemulsion-based flowback aid that delivered optimal fluid compatibility at elevated temperature, the most favorable emulsion-break profile, and the highest rapid

mobility value. These results suggest that the surfactant blend is an ideal candidate for evaluation in a field deployment campaign that applies a solvent-based stimulation fluid system for gas-condensate cleanup.

Testing the flowback-aid-laden stimulation fluid system in the presence of well-specific formulation materials, such as cuttings, crude oil, and brine, revealed a treatment formulation that achieved the lowest IFT, promoted rapid hydrocarbon flow through sandpack columns, and exhibited the highest flash point. These results establish the system as well-suited for field deployment.

Field implementation of the laboratory-informed stimulation fluid System 1-B delivered a combined production uplift of approximately 2,500 BOPD in the five wells tested. Each well demonstrated substantial and sustained production increases after the intervention.

The integration of micro-emulsion surfactants into solvent-based stimulation fluids represents a technically and economically viable strategy for remediation of condensate banking in gas and condensate reservoirs where conventional acidizing techniques have limited effectiveness.

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